Retail Shopping Behaviour Visualization Report

Introduction to Visual Data Analytics

DV1699 - 2026

Project Assignment

Group 7

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1 Introduction

This project analyzes a retail online shopping dataset through visual data analytics. The aim is to design visualizations that help retail users understand customer shopping behaviour, promo code usage, customer satisfaction, and sales performance across product categories and locations. The design applies course concepts such as data and task abstraction, marks and channels, visual encoding, color theory, perception, interaction, and ethical visualization design.

2 Dataset and Preparation

2.1 Dataset

The dataset was obtained from Kaggle and contains approximately 3,900 customer shopping records. Each record represents one purchase and includes categorical attributes such as gender, product category, location, promo code usage, and payment method, as well as quantitative attributes such as age, purchase amount, review rating, discount value, and previous purchases. After cleaning, the dataset contained 3,625 records and 12 columns.

Table 1 Data cleaning actions

Data Issue	Column	Cleaning Action	Reason
Missing values	Gender	Filled missing values with **Unidentified**	To keep the records while marking the gender as unidentified
	Discount Value	Filled missing values with **0**	To indicate that no discount was applied
Invalid values	Age	Removed rows with age value **0**	Age 0 is not valid for customer analysis
Inconsistent values	Gender	Replaced **Female** with **F**, **Male** with **M**, and **Unidentified** with **U**	To standardize gender categories
	Previous Purchases	Replaced **“nothing”** with **0**	To make the column numeric and consistent
	Payment Method	Replaced **Debit** with **Debit Card**	To standardize payment method categories
Unclear column name	Discount value	Renamed **Discount value** to **Discount Value (%)**	To clarify that the values are percentages

Data Issue	Column	Cleaning Action	Reason
Unnecessary columns	Year Country	Removed **Year** and **Country**	These columns contained only one unique value and were not useful for analysis

3 Potential Users and Tasks

3.1 Potential Users

The visualizations are intended for retail users who need to explore customer behaviour, evaluate promotions, compare regional performance, and identify business risks.

Table 2 Potential users and analytical needs

User Group	Main analytical need
Store manager	Identify strong and weak product categories, regions, and customer satisfaction issues
Marketing analyst	Evaluate promo code usage, understand customer response across locations, and support targeted marketing strategies
Sales analyst	Compare purchase patterns across age groups, categories, and locations
Business decision maker	Obtain a quick overview of retail performance, customer satisfaction, and possible improvement areas

3.2 Tasks

The users identified in this project need to perform several important analytical tasks.

Table 3 User task mapping

User task	Store manager	Marketing analyst	Sales analyst	Business decision maker
Task 1: Analyze whether promo code usage is associated with purchase amount across age groups	-	√	√	√
Task 2: Identify states with high sales volume but low customer review ratings	√	-	√	√
Task 3: Compare customer satisfaction across locations	√	√	√	√
Task 4: Analyze sales performance across product categories and regions	√	√	√	√

4 Visualization Design and Theoretical Motivation

The design was guided by the user tasks and by visualization principles from the course literature. The expressiveness principle was used by matching data types with appropriate encodings, while the effectiveness principle was used by prioritizing effective channels such as position and length for important quantitative comparisons (Munzner, 2015). Color was used mainly for grouping, highlighting, and ordered quantitative values, while perception issues such as contrast, distinctness, and color vision deficiency were considered (Ware, 2012; Ware, 2022).

Table 4 Visualization design choices and theoretical motivation

Task	Visualization	Main encoding	Motivation	Theory Reference
Promo code usage and purchase amount across age groups	- Grouped bar chart - Boxplot	<u>Bar chart:</u> - X-axis = age group - Y axis = average purchase amount - Color = promo code usage <u>Boxplot</u>: - X-axis = promo code usage - Y-axis = purchase amount - Grouped by age group	- The grouped bar chart supports direct comparison of average purchase amount between promo and non-promo users across age groups. - The boxplot complements it by showing the distribution, median, spread, and possible outliers, so the analysis does not rely only on average values	Bar charts are suitable for categorical comparison, while boxplots are useful for comparing distributions. Position and length are effective visual channels for quantitative values (Munzner, 2015; Ware, 2012; Course lecture slides, 2026).
High sales volume and low review ratings by state	Bubble chart	- X-axis: average review rating - Y-axis: total sales - Size: sales volume - Color: rating level	- The bubble chart allows users to compare sales performance and customer satisfaction at the same time. States with high sales but lower ratings can be identified as priority risk areas. - To improve transparency, the visualization presents two charts. The first chart shows the full review rating scale from **1.0 to 5.0**, while the second chart zooms in to the observed range of **3.5 to 4.0**. This allows users to understand the true rating scale while still being able to compare small differences between states.	Bubble charts are suitable for relational analysis when comparing multiple quantitative variables. Position is used for the main variables, while size and color provide additional information (Munzner, 2015; Ware, 2012).
Customer satisfaction across locations	Choropleth map	- Area: state - Color intensity: average review rating - Hover: detailed values	A map is suitable because the data has geographical attributes. It helps users identify regional differences in customer satisfaction.	Spatial visualizations support geographical pattern analysis. Ordered color scales are suitable for quantitative attributes because they show progression from lower to higher values (Munzner, 2015; Ware, 2012; Course lecture slides, 2026).
Sales performance across product categories and regions	- Treemap - Grouped bar chart	- Treemap area: total sales - Color: product category - Bar length: total sales	The treemap gives an overview of part-to-whole contribution, while the bar chart provides more accurate comparison across states and categories.	Treemaps are suitable for overview and part-to-whole analysis, while bar charts are better for precise comparison because bar length is easier to compare than area (Munzner, 2015; Ware, 2012; Spence, 2014).

Table 5 Perception and color design considerations for each visualization

Task	Visualization	Perceptual factors considered	Design consideration
Task 1	Grouped bar chart for promo code usage and age group	Distinctness, contrast, number of colors, semantics	- Promo and non-promo groups are separated using two distinct colors. - The bars contrast clearly with the white background, and only two main colors are used to avoid visual clutter. - The same color meaning is kept throughout the chart, making comparison between age groups easier.
	Boxplot for purchase amount distribution	Distinctness, contrast, color vision deficiency, field size	- The boxplot uses the same color mapping as the grouped bar chart, so users do not need to relearn the meaning of the colors. - The grouped position, axis labels, and legend also support interpretation, so the chart does not rely only on color. - The chart size is large enough to show the distribution across age groups clearly.
Task 2	Bubble chart for high sales and low review ratings	Distinctness, contrast, field size, semantics	- The bubble chart uses position to show the two main quantitative variables: average review rating on the x-axis and total sales on the y-axis. - Bubble size represents sales volume, while color represents average review rating. - A yellow-to-green sequential color scale is used because review rating is an ordered numerical variable on a 1.0 to 5.0 scale. Lighter yellow represents lower average satisfaction, while greener colors represent higher average satisfaction. This softer color scale avoids making small rating differences appear too extreme.
Task 3	Choropleth map for customer satisfaction across locations	Ordered color scale, contrast, semantics, color vision deficiency	- The map uses the same ordered color scale as the bubble chart to represent average review rating across states. This is suitable because review rating is quantitative and ordered. - Since color differences can sometimes be difficult to interpret, hover tooltips are included to show the exact rating and sales values. - This design choice supports a more truthful interpretation and avoids exaggerating small differences between states.
Task 4	Treemap for sales by product category and region	Distinctness, number of colors, field size, semantics	- The treemap uses area size to show sales contribution and color hue to distinguish product categories. - The number of product categories is limited, which helps keep the chart readable. - The treemap is used mainly for overview because area is less precise than bar length for exact comparison.
	Grouped bar chart for sales performance	Distinctness, contrast, number of colors, field size	- The grouped bar chart uses bar length to support accurate comparison of sales values across states and product categories. - Product categories are separated using different colors. - The layout, legend, and axis labels help users compare categories clearly without relying only on color.

5 Visualization Results and Discussion

5.1 Task 1: Promo Code Usage and Purchase Amount Across Age Groups

The essential goal of this task is growth validation. This task evaluates whether promo code usage is visually associated with higher spending across age groups.

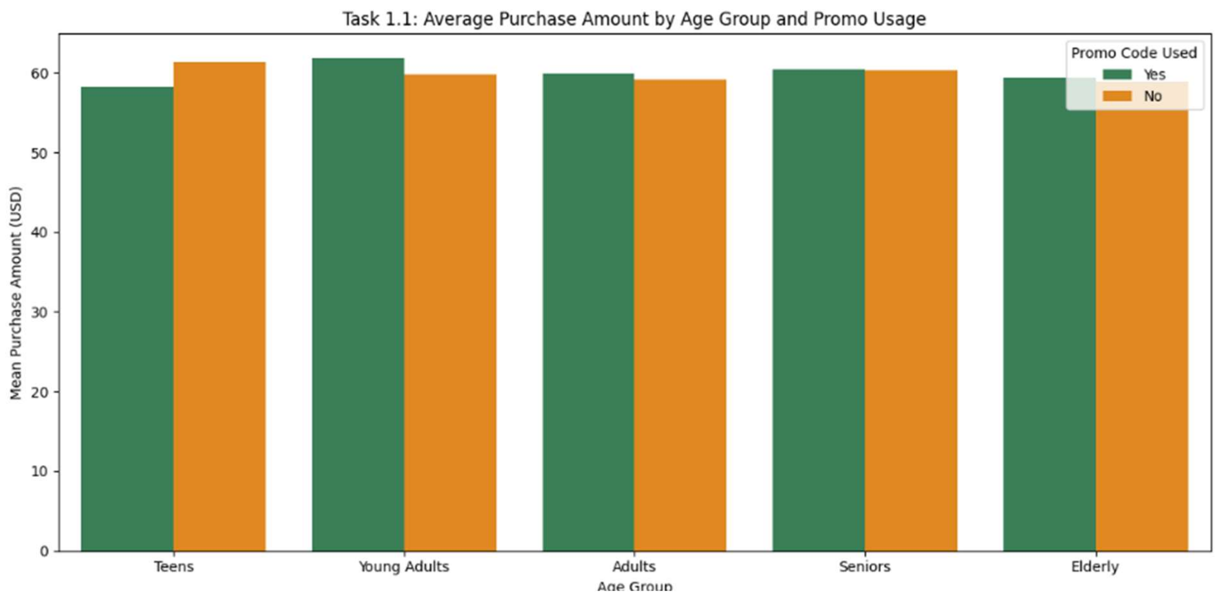

Figure 1 Average purchase amount by age group and promo usage

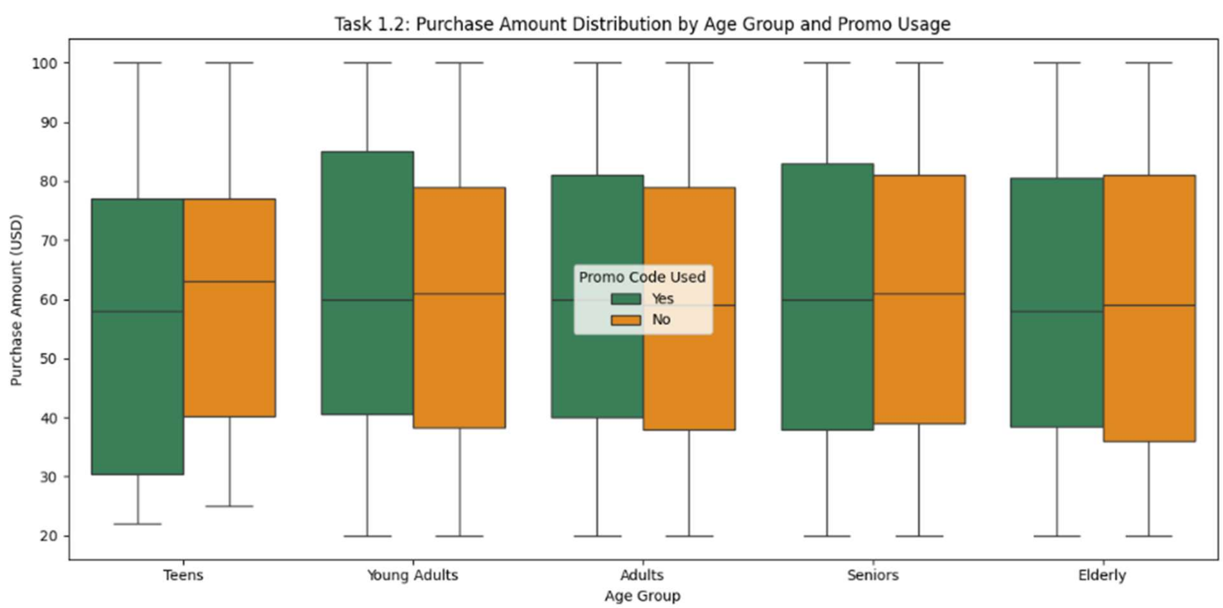

Figure 2 Purchase amount distribution by age group and promo usage

Interpretation:

Average purchase amounts are similar between promo-code users and non-promo users across age groups. The boxplot shows strong overlap in the purchase amount distributions, meaning the small differences shown in the bar chart should be interpreted carefully. There is no strong visual evidence that promo-code use is clearly associated with higher customer spending.

5.2 Task 2: High Sales Volume and Low Review Ratings by State

The essential goals of this task are risk management and profit protection. This task identifies states where strong revenue may hide customer satisfaction risks.

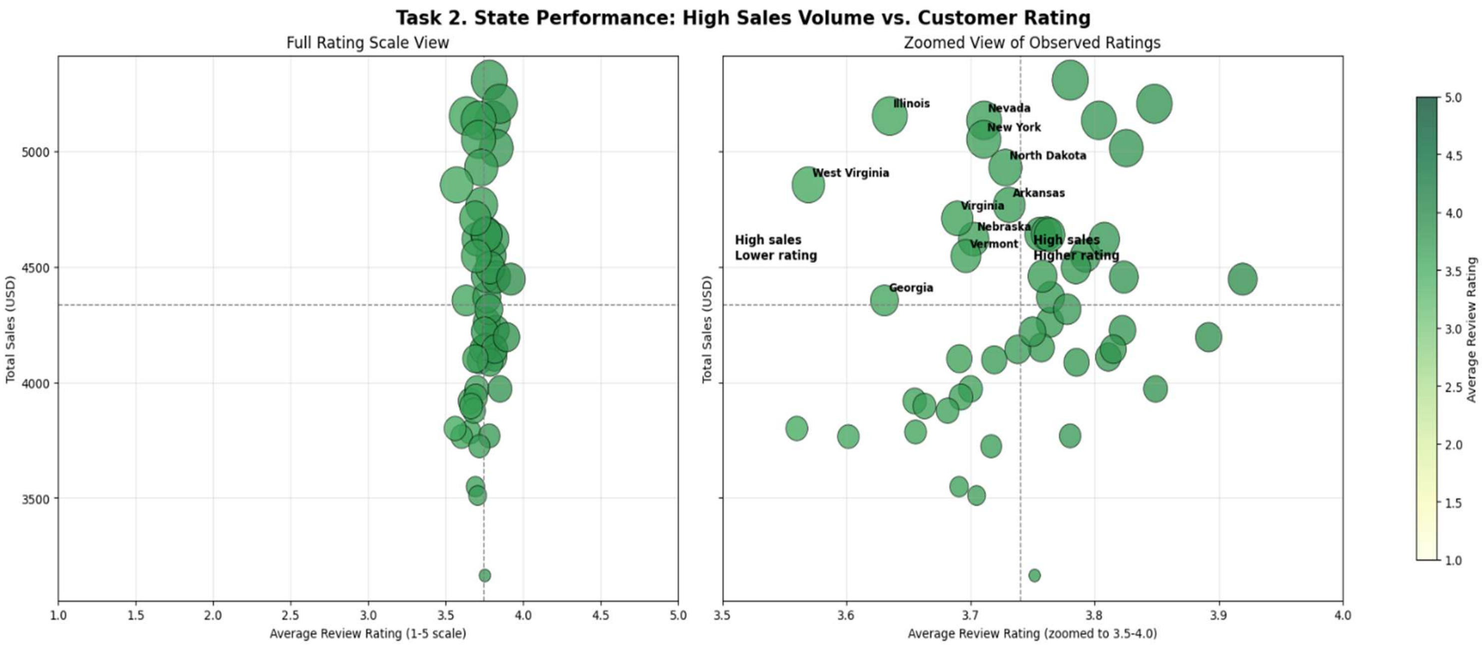

Figure 3 Total sales and average review rating by state

Interpretation:

- The two-chart design supports a more transparent interpretation.
- The full-scale chart shows that the differences in average review ratings are not extreme, while the zoomed chart helps users see the relative differences more clearly.
- The upper-left quadrant identifies states with higher average sales but lower average review ratings, including Illinois, Nevada, New York, North Dakota, and West Virginia. These states may require further investigation because high sales volume does not always indicate high customer satisfaction.
- Since the rating differences are small, this result should be interpreted as a priority signal rather than evidence of severe dissatisfaction.

5.3 Task 3: Customer Satisfaction Across Locations

The essential goal of this task is to examine whether customer satisfaction is consistent across locations. If ratings are low across most states, the problem may be related to the product itself. However, if ratings are high in some states and lower in others, the issue may be more local, such as delivery, service quality, or operational performance.

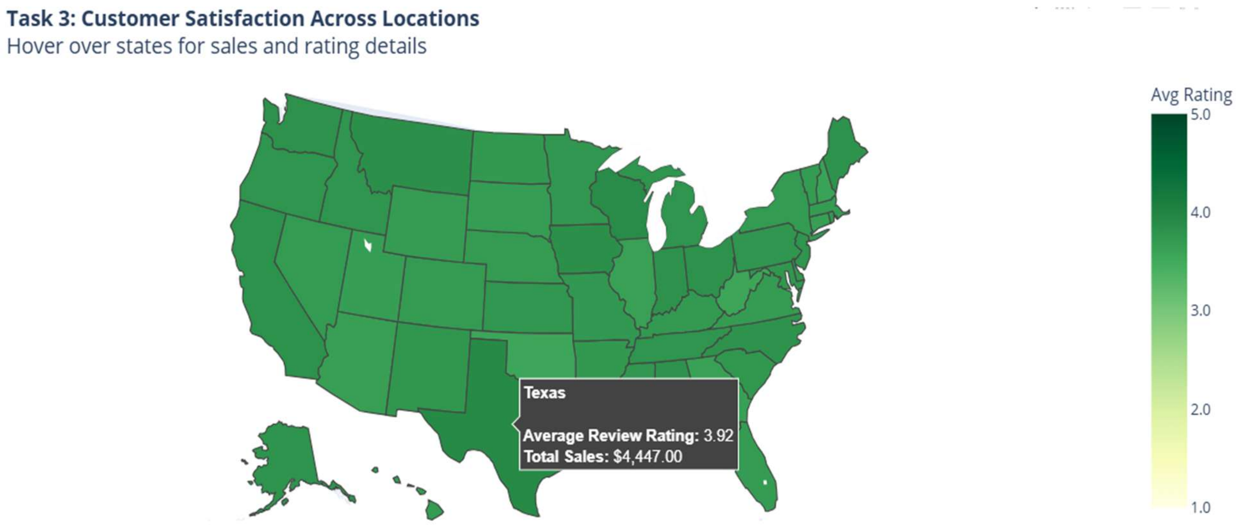

Figure 4 Choropleth map of average review rating by state

Interpretation:

Customer satisfaction varies across states, but the differences are relatively small. The map helps identify regional patterns in average review ratings, but it should not be interpreted as showing extreme gaps in customer satisfaction. The 1.0 to 5.0 color scale makes the visualization more transparent because it shows the state differences within the full original rating range. This helps avoid exaggerating small variations in average ratings.

5.4 Task 4: Sales Performance Across Product Categories and Regions

The essential goal of this task is to understand how sales are distributed across product categories and regions. This helps us see whether certain regions prefer certain product categories and which locations contribute the most to sales. This can help the business make better decisions about inventory, marketing, and regional sales strategy.

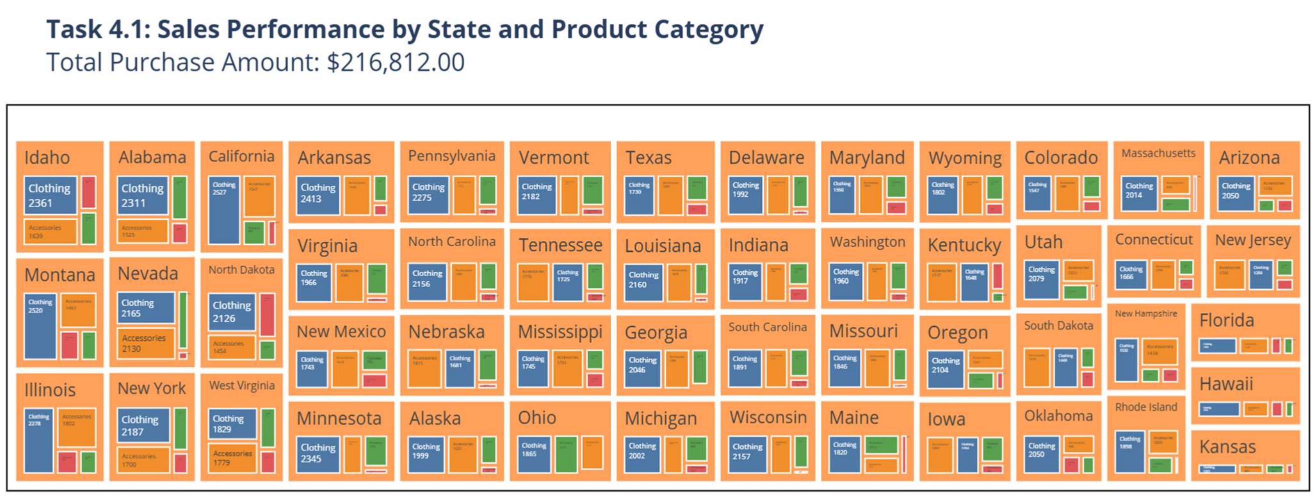

Figure 5 Sales performance by state and product category

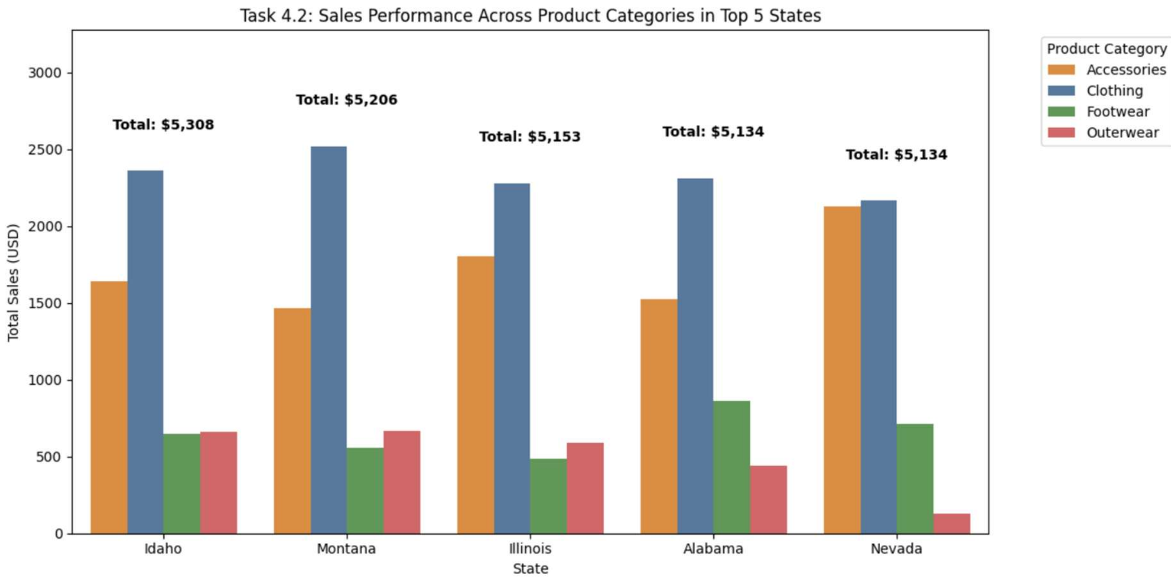

Figure 6 Product category sales in the top five states by total sales

Interpretation:

1. The treemap provides an overview of sales contribution across all states and product categories. Larger rectangles represent higher sales values, making the chart useful for quickly identifying which states or categories contribute more to overall sales. However, the treemap is mainly used for overview rather than precise ranking, because comparing rectangle areas is less accurate than comparing bar lengths.
2. The grouped bar chart focuses on the top five states by total sales: Idaho, Montana, Illinois, Alabama, and Nevada. These states have relatively similar total sales values, so no single state clearly dominates the top group. Across these states, Clothing is usually the strongest product category, followed by Accessories in several cases. This suggests that Clothing and Accessories are important contributors to sales performance in the highest-sales states.
3. These insights from both charts can support business decisions related to inventory planning, regional marketing, and product promotion.

6 Ethical Considerations

Ethical aspects of data visualization were considered during the design of the visualizations to avoid misleading interpretation and to support fair, clear, and honest communication of the data.

The main ethical focus was to present the dataset truthfully, avoid exaggerating small differences, and make the visualizations understandable for users.

Table 6 Ethical considerations in the visualization design

Ethical aspect	How it was considered in this project
Avoiding causal claims	The report uses careful wording such as “associated with” instead of “causes” or “increases.” For example, the promo code analysis does not claim that promo codes directly increase spending, because the visualization only shows patterns, not causality.
Avoiding exaggeration	- Small differences in review ratings across locations are not described as extreme. - The customer satisfaction map and bubble chart are interpreted carefully because the rating differences between states are relatively small.
Truthful color use	Strong or alarming colors are avoided when the numerical differences are small. For example, the customer satisfaction visualizations use an ordered color scale with a fixed 1.0 to 5.0 rating range so users do not overestimate small differences between states.
Appropriate visual encoding	- Bar charts are used when accurate comparison is important because bar length is easier to compare than area. - Treemaps are used mainly for overview because comparing area is less precise for exact ranking.
Axis transparency	- Review rating is originally measured on a 1.0 to 5.0 scale. - For the bubble chart, two views are provided: one chart shows the full 1.0 to 5.0 scale, while the second chart zooms in to the observed range of 3.5 to 4.0. This makes small differences easier to compare while still showing the original scale transparently.
Accessibility and readability	The charts do not rely only on color. Axis labels, legends, positioning, chart titles, and category labels are also used to support interpretation.
Transparency about limitations	The report explains that the dataset can reveal patterns, but it cannot fully explain the causes behind them. Some findings require further investigation before business decisions are made.
Avoiding visual clutter	- The number of colors and visual elements is limited to keep the charts readable and reduce cognitive overload. - The treemap is used for overview, while the grouped bar chart is used for clearer comparison of selected top states.

7 Summary of Main Findings

The main findings from the visualizations are summarized in Table 7.

Table 7 Main findings

Task	Main finding
Promo code usage	There is no strong visual evidence that promo-code use is clearly associated with higher customer spending.
High sales and low ratings	Some high-sales states have below-average review ratings and should be investigated further. However, the rating differences are small, so this should be interpreted as a priority signal rather than severe dissatisfaction.
Customer satisfaction by location	Customer satisfaction varies by state, but the differences are relatively small and should not be interpreted as extreme gaps.
Sales by category and region	Clothing and Accessories are the strongest contributors to sales performance, especially in the highest-sales states.

8 Conclusion

This project shows how visual data analytics can transform retail transaction data into meaningful insights for decision making. The visualizations helped identify important patterns related to promo-code use, customer satisfaction across locations, high-sales states with potential satisfaction risks, and sales performance across product categories.

Overall, the findings suggest that promo-code use is not clearly associated with higher customer spending, while some high-sales states may still require further investigation because strong sales do not always indicate high customer satisfaction. The analysis also shows that customer satisfaction differences between states are relatively small and should be interpreted carefully. In terms of product performance, Clothing and Accessories appear to be the strongest contributors to sales, especially in the highest-sales states.

Ethical visualization principles were considered throughout the project by avoiding causal claims, avoiding exaggeration, using appropriate visual encodings, and explaining limitations clearly. The visualizations support better business understanding by identifying sales patterns, customer satisfaction differences, and areas that may require further investigation.

9 References

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